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Analysing the correlation & creating a predictive model for determining
the detrimental effects digital consumption patterns have on refractive
errors & vision health

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SARAJEVO
2026

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SUMMARY

The proliferation of digital devices in contemporary life has coincided with an unprecedented rise in the prevalence of refractive errors and digital eye strain affecting younger populations. Existing literature indicates influence of digital factors on refractive errors, including total daily screen time and continuous screen exposure duration among others. Alongside these, non-digital factors such as age and genetic predisposition have been independently established as significant contributors to refractive error development. A frequently co-occurring condition, Computer Vision Syndrome (CVS), characterised by symptoms including eye strain, headaches, and nausea, has similarly been linked to prolonged digital device usage and the presence of pre-existing refractive errors. However, a gap remains in the literature concerning the compounding effects these variables exert upon one another, as well as the predictive applicability of such findings. This study aims to address these gaps by extending the current understanding of how digital consumption patterns collectively influence refractive error progression and overall vision health. To overcome limitations imposed by data scarcity in the healthcare and digital consumption domain, attributed largely to privacy conventions, this research employs proven data augmentation techniques, specifically Synthetic Minority Over-sampling Technique, to generate a synthetic yet statistically grounded dataset. Building upon this, a predictive machine learning model will be developed and trained to classify individuals into risk categories based on their digital usage profiles. This study creates a novel methodological comparison between factor weights derived from existing literature and those proposed by the machine learning model, offering insight into the degree to which empirical research aligns with data-driven predictions.

Keywords: machine learning, refractive errors, digital consumption, computer vision syndrome, data augmentation, predictive modelling

INTRODUCTION

1.1. A Transition From Analog to Digital

The shift from analog to digital information consumption represents a fundamental transformation in the nature of visual stimuli processed by the human ocular system. Unlike reflective print media, modern emissive digital displays operate on fundamentally different optical principles, exposing the visual system to high-frequency, near-field signals for which it was not evolutionarily optimised. From a systems perspective, this creates a condition in which the biological hardware — the human eye — is increasingly required to interface with a technological environment that exceeds its native operational parameters. This study addresses this incompatibility by treating ocular deterioration not merely as a clinical condition, but as a complex, multifactorial system response that can be modelled and predicted through digital consumption patterns and environmental variables, grounded in data derived from current empirical research.

A secondary but equally consequential dimension of this transition concerns the nature of visual engagement itself. Contemporary occupational and recreational contexts increasingly demand sustained, high-intensity focus on near-field digital screens, whether laptops, desktop monitors, or mobile devices. The compounding effect of prolonged near-work demand and the inherent optical properties of emissive displays creates conditions conducive to both acute visual discomfort and long-term refractive error progression.

1.2. Myopia: The most significant error

Within the scope of this research, myopia is conceptualised as a structural output error characterised by axial elongation of the eye, whereby environmental cues — most notably sustained near-work — function as high-weight input variables that

stimulate excessive ocular growth. This framing is supported by current meta-analytic evidence demonstrating a dose-response relationship between daily screen time and myopia risk, wherein each additional hour of daily screen exposure is associated with a 21% increase in myopia risk (Ha et al., 2025), with this effect exhibiting a sigmoidal pattern that plateaus beyond approximately four hours of daily usage. This threshold is of particular methodological relevance, as it informs the incorporation of non-linear constraints within the machine learning architecture employed in this study, wherein the effect of screen time beyond the four-hour mark is attenuated but not disregarded. Additionally, the model accounts for the absence of outdoor light exposure as a compounding risk variable, reflecting the established role of retinal dopamine in inhibiting excessive axial elongation.

1.3. Temporary effects: Computer Vision Syndrome

Beyond the long-term structural implications of digital screen exposure, this study accounts for acute and recurring visual strain through the lens of Computer Vision Syndrome (CVS). Emissive display technologies, including LCD and LED panels, introduce specific forms of optical noise: the pixel structure of digital screens, characterised by soft Gaussian edges, places the ciliary muscles in a state of continuous accommodative demand, contributing to cumulative eye-strain. A well-documented physiological consequence of this is a marked reduction in blink rate, which has been observed to decline from an average of approximately 20 blinks per minute to as few as 5 blinks per minute during sustained screen use (Swetha et al., 2025). Environmental technical variables, including Pulse Width Modulation (PWM) flickering and blue-light emission inherent to LCD and OLED display technologies, are further incorporated as noise factors within the study's analytical framework. As noted by Żuczek et al. (2025), these variables disrupt circadian rhythm regulation and melatonin production, generating a feedback loop with

implications for both long-term refractive error development, specifically myopia (Zhang et al., 2026), and the frequency with which individuals experience CVS-related symptoms, including dry eyes, nausea, and headaches.

1.4. The overall goal of this study: Data & Prediction

Despite the substantial body of literature examining individual risk factors for ocular deterioration, a meaningful gap persists in the capacity to predict personal ocular risk through a holistic, integrative assessment of digital habits, biological predispositions, and technical environmental variables. Existing research predominantly examines factors such as screen time, display characteristics, and genetic predisposition in isolation, thereby failing to capture the compounding and interactive dynamics that govern real-world refractive error progression. This study is motivated by the need to bridge this gap through the quantification of inter-factor relationships and the systematic modelling of each variable's relative contribution to refractive error risk. Key predictors identified in the literature, including continuous screen usage duration — heavily weighted as a driver of CVS symptom severity (Ece & Altındaş, 2025) — and visual task intensity, identified as a significant contributor to myopic progression (Ha et al., 2025), serve as foundational inputs to this model.

The core objective of this research is to conduct a multifactorial analysis of the relationship between digital consumption patterns and refractive errors, with primary emphasis on myopia, alongside secondary consideration of astigmatism and CVS as frequently co-occurring conditions. To address the data scarcity inherent to the healthcare and digital consumption domain, this study employs Synthetic Minority Over-sampling Technique (SMOTE) to generate a statistically grounded synthetic dataset calibrated to established weighted probabilities derived from the literature. A predictive model will subsequently be trained using XGBoost and Random Forest algorithms, yielding a baseline ocular risk index capable of identifying high-risk

individuals and providing actionable intervention thresholds. A novel contribution of this study lies in the systematic comparison between literature-derived factor weights and those produced by the machine learning model, offering a rigorous assessment of the degree to which existing empirical evidence aligns with data-driven predictions — and where further investigation may be warranted.

LITERATURE REVIEW

2.1. Historical Review of the Topic

Refractive errors have been a longstanding subject of scientific inquiry, with early research predominantly centred on genetic predisposition as the primary determinant of ocular development. While genetic factors remain a relevant and well-supported contributor in contemporary research, the prevailing consensus has progressively shifted toward a more comprehensive, multifactorial framework that accounts for the complex interplay between hereditary predisposition and environmental influences — most notably, the role of excessive and prolonged digital consumption.

From a biological standpoint, the human eye functions as a precision optical instrument in which the cornea and crystalline lens work in concert to focus incoming light precisely onto the retina. When a mismatch arises between the axial length of the eye and its refractive power, light fails to converge at the correct focal point, resulting in refractive error. Refractive errors affect a substantial proportion of the global population, a significant share of which remains uncorrected, rendering them the leading cause of vision impairment worldwide (Harb & Wildsoet, 2019). The proliferation of screen-dependent occupations and the steady rise in average daily screen time have compounded this burden considerably. Carr and Stell (2020) propose that prolonged near-work induces a hyperopic blur on the retina as a consequence of inadequate accommodation, which in turn signals the eyeball to structurally elongate in an attempt to restore focus — a mechanism understood to drive long-term refractive error progression.

A significant contribution to the quantification of this environmental impact is the 2025 systematic review and dose-response meta-analysis conducted by Ha et al.

(2025), encompassing over 335,000 participants. This study identified a sigmoidal dose-response relationship between daily digital screen time and myopia risk, wherein each additional hour of screen exposure corresponds to a 21% increase in the odds of myopia development (Odds Ratio: 1.21). The sigmoidal nature of this curve implies a relative safety threshold below one hour of daily exposure, with risk plateauing beyond four hours as biological susceptibility approaches saturation — though incremental damage continues beyond this threshold, albeit at a diminishing rate. This non-linear relationship has direct implications for the modelling approach adopted in this study, as it necessitates the incorporation of non-linear constraints within the machine learning architecture to accurately capture this dose-response dynamic. The risk profile associated with screen exposure is further compounded by multi-device usage; the concurrent or alternating use of smartphones and computers has been shown to increase myopia odds, likely attributable to the varied focal distances and the heightened accommodative demand involved in transitioning between devices (Foreman et al., 2021). It is important to note, however, that the Ha et al. (2025) analysis examines screen time as an isolated variable, and does not account for the co-variation and cross-effects of other environmental factors — such as profession, continuous screen interval duration, or ambient lighting — on refractive error development. This limitation underscores the necessity of an integrative modelling approach, as pursued in the present study.

The role of circadian rhythm disruption in refractive error progression represents an emerging and increasingly relevant dimension of this field. Prolonged exposure to blue-light emitting displays has been linked to suppression of melatonin production and disruption of natural sleep-wake cycles, with downstream implications for ocular health. Zhang et al. (2026) attempt to model this association through animal studies, confirming a cross-effect between disrupted sleep patterns, melatonin dysregulation, and myopic progression. While the findings are indicative, the exclusive reliance on animal models limits direct extrapolation to human populations, and the precise weight attributable to circadian-related factors — including the use of blue-light filters and night mode settings — in human refractive error development remains an open empirical question.

Regarding Computer Vision Syndrome (CVS), it is important to distinguish this condition from structural refractive errors, as CVS represents a constellation of transient, strain-induced symptoms rather than a permanent morphological change. A comprehensive review by Kaur et al. (2022) identifies the primary symptoms of CVS — also referred to in the literature as Digital Eye Strain — as including dry eyes, itching, foreign body sensation, excessive lacrimation, blurred vision, and headache. The COVID-19 pandemic served as an inadvertent natural experiment in this regard, with enforced screen dependency during lockdown periods associated with a 50% increase in CVS symptom prevalence among children alone, attributable to the abrupt substitution of outdoor activity with screen-based learning and communication. Swetha et al. (2025) elaborate further, noting that sustained near-work on digital devices induces continuous accommodative and convergence demand, which manifests as ocular fatigue and the symptomatic discomfort characteristic of CVS. At the physiological level, this presents as significantly reduced blink rates, elevated cognitive load, and evaporative tear film stress. Preventive strategies documented in the literature include ergonomic screen positioning, adherence to structured break schedules — most notably the 20-20-20 rule — and the prophylactic use of lubricating eye drops. A cross-sectional study by Raziya et al. (2022), conducted across 100 participants using a structured questionnaire and camera-based blink rate assessment, found that higher screen exposure in adolescents was significantly associated with reduced blink rates and greater CVS symptom burden, with behavioural interventions promoting conscious blinking and regular breaks identified as effective mitigation strategies. The authors themselves acknowledge, however, that the modest sample size and self-reported nature of the data limit the generalisability of these findings — a limitation that the synthetic data generation approach employed in the present study is specifically designed to address.

The type of display technology an individual uses has emerged as an increasingly important variable, exerting influence on both CVS through direct visual modification and myopia through indirect pathways including eye fatigue and circadian disruption. Among the most prevalent display technologies, LCD and OLED panels — both of which are emissive in nature — are generally considered

the most ocularly taxing under extended use, primarily due to their emission of short-wavelength blue light, which has been demonstrated to disrupt circadian rhythms and contribute directly to eye strain (Żuczek et al., 2025). In contrast, E-Ink display technology, which operates on a reflective rather than emissive principle by utilising ambient light rather than a self-powered light source, has gained traction as a more ocularly conservative alternative. While E-Ink displays substantially reduce the eye strain associated with emissive technologies, they do not eliminate the near-work demand inherent to prolonged screen reading and therefore do not offer a comprehensive reduction in myopia risk or CVS symptom frequency. Additionally, technical characteristics such as Pulse Width Modulation (PWM) flickering — a method used to regulate screen brightness in many LCD and OLED displays — introduce a form of imperceptible but physiologically relevant visual noise that contributes to cumulative ocular fatigue, particularly during extended usage sessions.

With respect to other refractive errors, the literature suggests a markedly differentiated risk profile when compared to myopia. Hyperopia and presbyopia, for instance, are predominantly governed by genetic predisposition and age-related biological changes respectively — specifically, the natural reduction in crystalline lens flexibility associated with ageing — rendering digital consumption a comparatively negligible factor in their aetiology (Harb & Wildsoet, 2019). Astigmatism, however, presents a more nuanced case. A study by Shah et al. (2025), conducted on a sample of 431 children with a mean age of 6.70 ± 1.80 years, identified a statistically significant association between increased screen time and astigmatism development through regression analysis, suggesting that digital consumption factors warrant consideration in astigmatism risk modelling, albeit with lower predictive weight than is attributed to myopia. On the basis of this evidence, hyperopia and presbyopia are excluded from the predictive scope of the present study, while astigmatism is retained as a secondary outcome variable.

2.2. The Basis for Predictive Modeling

Recent research has seen a meaningful transition toward quantitative risk assessment leveraging machine learning algorithms, driven by the increasing availability of

large-scale clinical datasets and advances in ensemble-based modelling techniques. The motivation for adopting a machine learning approach within the present study stems from the demonstrated capacity of algorithms such as Random Forest and XGBoost to capture complex, non-linear relationships between multiple co-varying predictors — a property that renders them particularly well-suited to the multifactorial nature of refractive error risk.

The Anyang Childhood Eye Study (Li et al., 2022) provides a notable precedent in this domain, employing both Random Forest and XGBoost algorithms to predict myopia progression in a longitudinal clinical cohort. The study identified Uncorrected Distance Visual Acuity (UDVA) as the most influential predictor, carrying a weighted importance of 28%, followed by Spherical Equivalent (SE) at 20%. Whilst these clinical models demonstrate high predictive accuracy, they are constrained by their reliance on longitudinal clinical measurements that are neither accessible nor replicable within a digital consumption research context. A significant and underexplored gap therefore persists: existing predictive models do not integrate granular, real-time digital consumption patterns — such as display type, usage intensity, continuous interval duration, and multi-device behaviour — nor do they jointly model these alongside non-digital factors such as profession, outdoor activity, and age, in relation to refractive error progression and CVS symptom frequency.

A further challenge inherent to this domain is the scarcity of suitable real-world datasets. Comprehensive data linking individual digital consumption behaviours to ocular health outcomes is exceptionally difficult to obtain at scale, primarily due to privacy constraints and the logistical complexity of longitudinal data collection in non-clinical settings. To address this, the present study constructs a synthetic, weighted, and factorised dataset grounded in empirically established dose-response relationships and literature-derived factor weights, designed to represent the population of digitally active individuals at potential risk of refractive error progression. While the use of fully synthetic datasets carries inherent limitations regarding real-world representativeness, Goyal and Mahmoud (2024) affirm that synthetic data generation offers a credible and increasingly adopted solution to the challenges of data scarcity, privacy preservation, and algorithmic bias mitigation in

machine learning research.

The data strategy employed in this study adopts a hybrid approach, in which a structured survey serves as the empirical foundation for data collection, with Synthetic Minority Over-sampling Technique (SMOTE) — and potentially Generative Adversarial Networks (GANs) in subsequent iterations — applied as the primary mechanism for dataset augmentation. By grounding the synthetic dataset in established dose-response probabilities and literature-validated factor weights, the model is trained within a statistically coherent framework while simultaneously circumventing the data accessibility constraints characteristic of healthcare and digital consumption research. This approach diverges from traditional cross-sectional survey methodology and establishes the foundation for a proactive predictive tool capable of identifying high-risk individuals prior to the onset of irreversible ocular damage. As cautioned by Hao et al. (2024), the use of synthetic data must be approached with transparency and methodological rigour; nonetheless, within the constraints of this research context — including the accessibility limitations inherent to student-led and smaller-scale academic projects — it represents the most scientifically defensible available method of data collation. Future research should seek to validate and refine the findings of this study through integration with real-world clinical and population-level datasets.

RESEARCH QUESTIONS / HYPOTHESES

With the literature review completed and its key findings thoroughly examined, it is necessary to explicitly delineate the gaps and contributions this research aims to address — either by providing substantive answers or by establishing a rigorous foundation upon which future research may build. Broadly, the scope of this paper is oriented toward determining the nature and magnitude of detrimental effects on ocular health attributable to digital consumption patterns, and toward developing both analytical and predictive frameworks capable of modelling these effects in an integrated, multifactorial manner.

3.1. Research Question One - Refractive Worsening

The relationship between digital consumption and refractive error development is an area in which a meaningful body of evidence already exists. Myopia and astigmatism have been demonstrated to be directly influenced by increased screen time, with Ha et al. (2025) quantifying a statistically significant increase in the probability of developing or experiencing worsening myopia with each additional hour of daily screen exposure. Complementary factors — including the duration of uninterrupted screen sessions and the viewing distance maintained between the eyes and the screen — have similarly been identified as contributors to refractive error risk. Critically, however, these factors do not operate with uniform effect; each carries a distinct directional and magnitude-based weight, with some functioning as risk-amplifying variables and others as protective or mitigating ones. Furthermore, many of these factors exhibit meaningful interaction effects with one another, such that the detrimental influence of one variable may be compounded by the simultaneous presence of another. For instance, an individual employed in a screen-intensive profession such as software development who also carries a genetic predisposition to

myopia through parental history will face a materially elevated risk profile compared to a peer in the same profession without such hereditary background.

It is within this context that the first research question and its associated hypothesis are proposed:

- **RQ1:** Can the individual effects, as well as the combined and weighted influence of digital consumption factors — including screen time, viewing distance, continuous usage interval duration, and intensity of visual focus — be used to predict the likelihood of an individual experiencing the onset or worsening of myopia and astigmatism?
- **H1:** Digital consumption factors, including screen time and intensity of usage, will predict the likelihood of an individual experiencing refractive error progression with a minimum average model accuracy of 75%.

3.2. Research Question Two - Frequency of CVS Experience

Alongside the investigation of long-term structural refractive changes, this research addresses the transient but clinically relevant phenomenon of Computer Vision Syndrome (CVS). As established in the literature review, CVS does not constitute a permanent structural impairment but rather a recurring cluster of discomfort symptoms — including eye strain, headaches, nausea, and dry eyes — that are directly and consistently associated with digital consumption patterns. The distinction between CVS and refractive errors is important from a modelling perspective: whereas the outcome of interest for RQ1 is the likelihood of structural ocular deterioration, the outcome of interest for RQ2 is the frequency with which an individual experiences CVS symptom episodes. The literature supports strong associations between CVS frequency and factors including high-intensity visual focus, rapid switching of gaze between multiple devices, display type, and ambient lighting contrast, among others.

On this basis, the second research question and hypothesis are defined as follows:

- **RQ2:** Can the individual effects, as well as the combined and weighted influence of digital consumption factors — including screen time, viewing distance, intensity of visual focus, and device usage patterns — be used to

accurately predict the frequency with which an individual experiences Computer Vision Syndrome symptoms, including headaches, eye strain, and visual fatigue?

- **H2:** Digital consumption factors, including screen time and intensity of usage, will predict the likelihood of an individual experiencing symptoms related to CVS (headaches, eye-strain and similar) with a minimum average model accuracy of 75%.

3.3. Analytical Objectives and Methodological Contributions

Beyond the two primary research questions outlined above, this study pursues several interconnected analytical objectives that collectively constitute its broader methodological contribution to the field.

The first of these is the development of a literature-grounded linear prediction framework. Drawing on the factor weights established and reviewed throughout the literature review, a set of linear equations will be manually constructed for each of the three outcome variables — myopia and astigmatism risk, as well as CVS frequency. Each equation assigns literature-derived weights on a normalised 0–1 scale to the relevant input factors, producing a continuous risk score that classifies individuals as either high-risk or low-risk based on their survey responses. This linear model serves not merely as a predictive tool, but as a structured, transparent representation of what the existing body of empirical research collectively suggests about the relative importance of each factor.

The second analytical objective is the development and training of a machine learning predictive model. Using XGBoost and Random Forest algorithms (or any other adequate algorithm) — selected for their demonstrated efficiency in handling non-linear, multi-variable classification tasks within health-related research contexts — the model will be trained on a synthetic, augmented dataset to generate its own data-driven weighting of each input factor. The dataset itself will be constructed through a hybrid approach: a structured self-reported survey will serve as the empirical foundation, with Synthetic Minority Over-sampling Technique (SMOTE)

applied to augment and balance the dataset in accordance with established dose-response probabilities and literature-validated distributions.

The third and most novel contribution of this study is the systematic comparison between literature-suggested weights and machine learning-derived weights across all input factors and outcome variables. Where the two sets of weights converge, this alignment serves as cross-methodological validation of the existing empirical literature. Where divergence is observed, the discrepancy will be interpreted as either an indication that the current literature requires more granular investigation, or that the synthetic dataset requires refinement through integration with real-world clinical data. In either case, this comparison provides a structured, evidence-based framework for identifying where the predictive understanding of digital consumption's effects on ocular health is robust, and where it remains incomplete.

Taken together, these objectives position this study not only as a predictive modelling exercise, but as a critical, integrative assessment of the current state of knowledge in this domain — one designed to be extended and built upon by future research with access to larger, real-world datasets.

DATASET

4.1. Data Context and Accessibility Constraints

A foundational consideration in the design of this research is the context under which the dataset was curated, as this directly informs both the methodological decisions made and the limitations that must be acknowledged by any subsequent research seeking to build upon these findings. Eyesight and ocular health data falls unambiguously within the protected category of healthcare information, and as such, existing datasets — whether maintained by large clinical organisations or produced as outputs of prior research — are either inaccessible to the general public or restricted to authorised institutions and individuals operating under formal ethical and legal frameworks. The fraction of ocular health data that is publicly accessible is either synthetic in nature, insufficiently granular, or incompatible with the specific structural requirements of this research. Rather than pursuing institutional data access — a process with a low probability of success within the scope and timeline of a student-led academic project — this study adopts a hybrid data strategy: a structured, purpose-built survey serves as the empirical foundation for real-world data collection, which is subsequently expanded through statistically grounded synthetic augmentation techniques.

4.2. The Case for Synthetic Data

While the credibility of synthetic datasets has historically been a subject of debate across disciplines, their adoption has accelerated considerably in parallel with the growth of machine learning research and the tightening of data privacy legislation globally. Synthetic data is increasingly recognised as a legitimate and effective mechanism for addressing real-world challenges of data scarcity, privacy

preservation, and algorithmic bias in machine learning applications. When the underlying real data is carefully curated and the augmentation process is methodologically sound, synthetically extended datasets have been demonstrated to preserve the statistical patterns and behavioural distributions of the original data while simultaneously increasing sample size and demographic diversity — without exposing any identifiable individual information (Goyal & Mahmoud, 2024). This approach carries particular relevance within the healthcare domain, where synthetic data generation has been adopted across a growing body of research not only to address privacy constraints but also to enhance the availability and diversity of training data for AI-driven diagnostic and predictive models, ultimately contributing to improved patient outcome modelling and broader healthcare research capacity (Pezoulas et al., 2024). It is important to emphasise, however, that the objective of the dataset constructed in this study is not to serve as a perfect real-world representation of the target population, but rather to function as a scientifically grounded training framework — one that captures the established correlational structure between digital consumption factors and ocular health outcomes, and upon which a predictive model can be reliably developed and evaluated. Full transparency regarding the survey structure, raw data, and augmentation procedures will be maintained within the final research output, in accordance with the principles of reproducibility and academic integrity, and to ensure that the limitations of this approach are clearly understood by those seeking to extend this work.

4.3. Data Augmentation: SMOTE

Given that the initial dataset produced by the survey is expected to be limited in size and unlikely to contain a balanced representation of all relevant outcome classes — particularly the minority cases of high-risk refractive error profiles — a structured augmentation strategy is required. For this purpose, the Synthetic Minority Over-sampling Technique (SMOTE) is employed as the primary method of dataset expansion. Rather than duplicating existing minority class instances — an approach that would introduce overfitting by failing to generalise beyond observed data points — SMOTE operates by identifying the feature-space neighbourhood of each minority sample and generating new, interpolated synthetic instances along the

vectors connecting a sample to its nearest neighbours (Chawla et al., 2002). The result is a dataset with a more balanced and continuous class distribution, enabling the predictive model to learn a broader and more representative range of outcome patterns.

Within the specific context of this study, SMOTE is applied not directly to the raw survey responses but rather to a set of aggregated numerical variables that represent grouped clusters of weighted input factors, each corresponding to a domain of influence on the target outcomes — myopia progression risk, astigmatism likelihood, and CVS symptom frequency. This design decision is motivated by three interconnected considerations. First, the volume of individual input factors is sufficiently large that direct augmentation at the raw feature level introduces unnecessary complexity; aggregating factors into domain clusters — such as a composite digital consumption score, a genetic predisposition score, and an environmental conditions score — simplifies the augmentation space without sacrificing the structural integrity of the underlying relationships. Second, raw survey responses are inherently discrete and fixed-step in nature, meaning that any synthetic values generated through interpolation would require rounding to valid response categories, introducing quantisation error that is avoided by operating on continuous aggregate scores. Third, given that this study models three distinct outcome variables — each of which assigns different relative weights to the same set of input factors — operating on pre-weighted aggregate scores allows the augmentation process to remain outcome-specific and avoids the conflation of factor weights across different prediction targets.

4.4. Future Augmentation: Generative Adversarial Networks

A prospective enhancement to the augmentation methodology outlined above involves the implementation of Generative Adversarial Networks (GANs). In contrast to SMOTE's reliance on linear interpolation between local feature-space neighbours, a GAN employs a deep learning architecture comprising two adversarially trained neural networks — a Generator and a Discriminator — which compete iteratively until the Generator produces synthetic samples indistinguishable

from real data (Goodfellow et al., 2014). This adversarial framework enables the model to learn the full, complex probability distribution underlying the dataset, capturing non-linear dependencies and higher-order interactions between digital consumption variables and ocular outcomes that geometric interpolation methods are structurally incapable of representing. The result is a synthetically generated population that preserves the statistical characteristics of the original survey data with substantially greater fidelity and diversity than SMOTE alone can achieve.

This approach is of particular relevance to the multi-target modelling framework of the present study. Specialised GAN architectures designed for tabular data — most notably the Conditional Tabular GAN (CTGAN) — are explicitly engineered to handle the mixed variable types and conditional dependencies characteristic of survey-based datasets, making them a well-suited candidate for generating synthetic populations that more robustly bridge the gap between simulated training data and real-world clinical distributions (Xu et al., 2019). While the implementation of GAN-based augmentation is positioned as a potential extension of the current work rather than a primary methodological commitment, its inclusion in this framework reflects the study's broader ambition to establish a scalable, extensible data infrastructure that can support increasingly sophisticated predictive modelling as access to real-world data improves.

4.5. A Note About the Dataset Structure

Although the case for synthetic data is already explained within the context of this paper, it is important to understand the underlying structure of the dataset itself. With the main data collection method being a questionnaire, the core structure of it is of vital importance for gathering the data. It is due to this reason that the survey itself is structured to capture two types of data points:

- Digital consumption & habits: These are the questions within the survey which describe the individual's nature regarding the intensity of their digital usage from various aspects, as well as other non-digital related factors such as their profession, outdoor activities... The consensus of these questions is to create a self-assessed image of the individual's digital habits. These are later

used within the context of the linear risk-assessment equation (with the weights of each factor calculated via their literature-based values) as well as within the context of the machine learning risk-assessment (with the weights of each factor adjusted via the ML's learning principles).

- Prescription information: This is a group of questions aimed to gather the predictive target. They capture data regarding the individual's current prescription (if possible their exact prescription in diopters, or at the very least the one they remember the last), any note of worsening refractive errors and the frequency of experiencing symptoms regarding CVS.

The self-assessed nature of the survey naturally contains a dose of self-bias and inaccuracy. However, this should be taken into account within the context of the paper and the generated dataset. Further research should aim to conduct a clinical study and data collection with in-person interviews and exact medical documentation.

METHODOLOGY

The methodology of this study adopts a quantitative, multi-phase technical workflow designed to transform self-reported digital consumption behaviours into objective, quantifiable risk scores across three ocular health outcome variables: myopia progression likelihood, astigmatism likelihood, and CVS symptom frequency. The methodological pipeline is structured across four sequential phases: (1) survey-based data collection, (2) factor weighting derived from the literature, (3) synthetic data augmentation via SMOTE, and (4) predictive model training, evaluation, and comparative analysis. Each phase is described in detail in the subsections that follow.

5.1. Data Sample Structure

Each individual data sample is structured as a questionnaire response comprising two categories of data points. The first category constitutes the input data, consisting of the factors identified through the literature review as the most empirically significant contributors to refractive error development and CVS symptom frequency. The second category constitutes the output data, representing the three target outcome variables against which the predictive model is trained and evaluated.

The input factors are not treated as uniform in their influence; each carries a distinct literature-derived weight that varies depending on the specific outcome variable being predicted. These weights, normalised on a continuous scale from 0.0 to 1.0, reflect the direction and magnitude of each factor's effect as established by the current body of empirical research. Where no precise quantitative specification exists in the literature, factor weights are assigned on the basis of the known biological mechanisms associated with that factor and the strength of indirect evidence available. The individual factors and their literature-derived weights are as follows:

- Average Daily Screen Time is one of the strongest predictors of CVS

symptoms ($w = 0.9$), given its direct causal relationship with eye strain, dryness, and headaches (Sheppard & Wolffsohn, 2018). It also carries substantial weight for myopia progression ($w = 0.7$), as sustained near-work demand from screens has been consistently linked to myopic shift (Huang et al., 2015). Its contribution to astigmatism is comparatively weaker ($w = 0.3$), with only limited evidence suggesting screen-related corneal shape changes (Hiraoka et al., 2018).

- Continuous Screen Usage Without Breaks is a strong predictor of CVS symptoms ($w = 0.85$), as uninterrupted near-work prevents the accommodative system from recovering, leading to eye strain and fatigue (Talens-Estarellles et al., 2023). This factor also holds moderate-to-high weight for myopia progression ($w = 0.65$), since prolonged uninterrupted near-work has been associated with sustained axial elongation of the eye (Lanca & Saw, 2020). Its effect on astigmatism remains limited ($w = 0.25$), with insufficient direct evidence linking break frequency to corneal shape changes (Wolffsohn et al., 2021).
- Intensity of Usage — wherein high-intensity tasks such as programming or reading require greater accommodative effort and reduced blink rates — constitutes a strong contributor to CVS symptoms ($w = 0.80$) relative to passive scrolling (Sheppard & Wolffsohn, 2021). For myopia, high cognitive demand tasks performed at near distances carry moderate-to-high weight ($w = 0.60$), as the increased accommodative load has been associated with greater myopic progression (Ghorbani Mojarad et al., 2021). The influence on astigmatism is weak ($w = 0.20$), as task intensity has not been meaningfully linked to corneal curvature changes in recent literature (Lanca & Saw, 2020).
- Viewing distance is a well-established risk factor for myopia progression ($w = 0.75$), as shorter working distances increase vergence-accommodation demand and have been directly linked to axial eye elongation (Lanca & Saw, 2020). It also carries significant weight for CVS ($w = 0.70$), since suboptimal screen distance forces the visual system to work harder, increasing strain and discomfort (Talens-Estarellles et al., 2023). Evidence for its role in

astigmatism development is limited ($w = 0.25$), though some studies suggest close viewing distances may exert mild biomechanical pressure on the cornea (Wolffsohn et al., 2021).

- Ambient Lighting Conditions — particularly poor contrast between screen brightness and surrounding light — constitute a moderate-to-strong contributor to CVS symptoms ($w = 0.75$), increasing visual discomfort and glare-induced strain (Sheppard & Wolffsohn, 2021). For myopia, dim indoor lighting carries moderate weight ($w = 0.55$), as low-light environments have been associated with increased accommodative lag and myopic progression, particularly in younger users (Ostrin et al., 2021). The link to astigmatism remains weak ($w = 0.20$), with no strong direct evidence connecting ambient lighting to corneal curvature irregularities (Lanca & Saw, 2020).
- Multiple Device Usage significantly elevates CVS risk ($w = 0.80$), as it increases total visual load, forces rapid refocusing between screens of varying sizes and distances, and further reduces blink rate (Talens-Estarellles et al., 2023). Its contribution to myopia progression is moderate ($w = 0.50$), since multi-device usage effectively extends near-work exposure and reduces the likelihood of adequate breaks (Lanca & Saw, 2020). Evidence for a direct link to astigmatism remains minimal ($w = 0.15$), as no recent studies have specifically isolated multi-device use as a corneal risk factor (Wolffsohn et al., 2021).
- Blue Light Filters and Night Modes have a moderate protective effect on CVS symptoms ($w = 0.50$), as reducing short-wavelength light exposure has been shown to decrease photophobic discomfort and improve perceived visual comfort during prolonged screen use (Lawrenson et al., 2021). Their role in myopia prevention is debated and assigned a low-to-moderate weight ($w = 0.35$), as clinical evidence for filter efficacy remains inconclusive (Lawrenson et al., 2021). Evidence for any meaningful effect on astigmatism is negligible ($w = 0.10$), as no credible recent literature links blue light filtering to changes in corneal shape or regularity (Sheppard & Wolffsohn, 2021).
- Profession carries moderate-to-high weight for CVS symptoms ($w = 0.75$), as

occupations requiring prolonged screen use consistently show elevated rates of digital eye strain and dry eye complaints (Sheppard & Wolffsohn, 2021). For myopia, profession holds moderate weight ($w = 0.55$), as higher occupational near-work demand has been repeatedly associated with greater myopia prevalence and progression in adults (Ghorbani Mojarad et al., 2021). Its contribution to astigmatism is low ($w = 0.25$), with findings regarding sustained near-work professions and corneal changes remaining inconsistent (Wolffsohn et al., 2021).

- Outdoor Activity is one of the most robustly supported protective factors against myopia progression ($w = 0.80$), with increased exposure to natural light stimulating retinal dopamine release that inhibits axial elongation (Xiong et al., 2020). Its protective weight for CVS is moderate ($w = 0.45$), as time spent outdoors naturally enforces breaks from near-work and allows the accommodative system to relax (Talens-Estarellles et al., 2023). Evidence for a direct protective role against astigmatism is limited ($w = 0.20$), with only indirect associations suggested through general eye health mechanisms (Lanca & Saw, 2020).
- Genetics is the single strongest predictor of myopia development ($w = 0.90$), with one or two myopic parents substantially elevating risk independently of environmental factors (Tedja et al., 2021). For astigmatism, genetic predisposition also carries high weight ($w = 0.80$), as corneal curvature and regularity are strongly heritable traits (Shah et al., 2021). Its direct weight for CVS is comparatively low ($w = 0.25$), as genetics may influence baseline tear film quality but does not directly drive symptom severity in most individuals (Wolffsohn et al., 2021).
- Age of first prescription is a strong factor for myopia ($w = 0.75$), as onset typically occurs during childhood and adolescence with progression stabilising in early adulthood (Lanca & Saw, 2020). For CVS, age carries moderate weight ($w = 0.60$), as older individuals experience reduced accommodative flexibility that amplifies digital eye strain under equivalent screen exposure (Sheppard & Wolffsohn, 2021). Its role in astigmatism is moderate ($w = 0.50$), as age-related changes in lens and corneal biomechanics

are well documented to alter astigmatic patterns over time (Wolffsohn et al., 2021).

- Gender carries a low-to-moderate weight for myopia ($w = 0.40$), with some studies identifying slightly higher myopia prevalence in females, though findings vary across populations and the effect is considered secondary to environmental factors (Rudnicka et al., 2023). For CVS, gender holds moderate weight ($w = 0.50$), as females have been consistently shown to report higher rates of dry eye and visual discomfort, likely linked to hormonal influences on tear film stability (Sheppard & Wolffsohn, 2021). Evidence for gender as an independent astigmatism risk factor is weak ($w = 0.20$), with no strong or consistent biological mechanism identified in recent literature (Shah et al., 2021).

The factors are subject to change as the research of the topic continues, however the bulk of the factors remains consistent with the above mentioned. To further facilitate compatibility with the synthetic augmentation process, the individual factors are aggregated into four composite domain groups, each representing a distinct category of influence on the target outcomes:

- Strain Index encompasses average daily screen time and continuous screen usage interval duration. These represent the highest-weight environmental predictors within the model and form the primary driver of risk score calculation across all three outcome variables.
- Digital Consumption Detriments encompasses viewing distance, intensity of usage, ambient lighting conditions, multiple device usage, and blue light filter usage. These factors carry moderate individual weights but exhibit meaningful interaction effects with the Strain Index — that is, their detrimental impact is amplified proportionally as the Strain Index of an individual increases.
- Lifestyle Effects encompasses outdoor activity and profession. These factors may exert either protective or risk-amplifying effects depending on the individual's profile, and are treated as directional modifiers within the composite scoring framework.
- Biological Factors encompass genetics, age, and gender. These represent

non-modifiable or slowly-changing individual characteristics that establish a baseline predisposition to refractive error development, independent of behavioural variables.

The full summary table of all factors, their group assignments, and their complete weight specifications across all three outcome variables is provided in Appendix A.

5.2. Data Collection Method - Survey

Primary data collection is conducted through a structured, purpose-built questionnaire designed to capture quantitative and ordinal-scaled data across the four factor domains identified above. The survey instrument employs a combination of Likert-scale response items and bounded numerical input fields, ensuring compatibility with the pre-processing and aggregation requirements of the synthetic augmentation pipeline. The use of structured surveys as a primary data collection instrument in ophthalmic epidemiology is well-established, with self-reported digital habit data consistently demonstrating meaningful correlations with clinical ocular discomfort and refractive outcomes (Rosenfield, 2011). Questions pertaining to the Biological Factors domain — particularly those concerning existing refractive prescriptions and parental myopia history — are phrased in alignment with standard clinical history-taking conventions to maximise response accuracy and minimise misclassification.

The target demographic for survey distribution encompasses individuals who are regularly and substantially exposed to digital devices through their education or occupation, with a primary focus on the age range of 6 to 30 years — the window in which myopia onset and progression are most clinically significant. Broad demographic variance within this target group is actively sought, as a diverse response pool is essential to ensure that the minority outcome classes are sufficiently represented prior to SMOTE augmentation. As noted by Dillman et al. (2014), the precision and quality of the initial data collection instrument directly conditions the validity of any downstream statistical modelling or machine learning inference; accordingly, significant care has been taken in the design of response categories to minimise ambiguity and reduce the likelihood of misreported behavioural data.

Several design principles have been applied to ensure the reliability and internal consistency of the survey instrument. First, all questions are framed in terms of typical or habitual behaviour rather than exceptional instances, reducing the influence of recall bias on response accuracy. Second, response options for screen time and continuous usage duration are provided as discrete interval ranges rather than open numerical fields, reducing the cognitive burden on respondents and improving the consistency of categorisation. Third, the survey is structured to progress from general digital consumption habits toward more specific biological and lifestyle factors, following an ordering convention designed to minimise priming effects between question domains. Fourth, the survey does not explicitly signal to respondents that eyesight risk is the subject of investigation, in order to reduce socially desirable response patterns that may otherwise skew the biological and lifestyle factor data.

In terms of sample size requirements, a minimum of 140 valid responses is established as the operational threshold for this study. Of these, 40 responses will be withheld as a held-out test set for final model evaluation, while the remaining 100 will constitute the training pool to which SMOTE augmentation is applied. This minimum threshold is further conditioned on the requirement that the minority outcome class within the training pool contains no fewer than 8 instances, as SMOTE requires a minimum neighbourhood size to generate interpolated samples without introducing excessive noise or sample duplication. It must be emphasised that these figures represent minimum operational requirements rather than targets; a larger and more demographically varied response pool would substantively improve the statistical robustness of the augmented dataset and the generalisability of the resulting model. The complete survey instrument, including all questions and response options, is provided in Appendix B.

Several limitations inherent to the survey-based data collection approach must be acknowledged. Self-reported behavioural data is subject to recall bias, social desirability bias, and imprecision in the estimation of habitual behaviours such as daily screen time — factors that may introduce systematic error into the input data. The voluntary nature of survey participation introduces a degree of selection bias, as

individuals with a personal interest in or awareness of eye health may be disproportionately represented among respondents. Furthermore, the survey captures a cross-sectional snapshot of respondents' current digital habits and biological characteristics, and is therefore unable to account for longitudinal changes in behaviour or refractive status over time. These limitations are partially mitigated by the synthetic augmentation strategy, which distributes the influence of individual data points across a broader synthetic population; however, they remain important constraints on the direct clinical applicability of the model's outputs and should be addressed in future research through longitudinal real-world data collection.

5.3. Predictive Modelling Framework

5.3.1. Linear Equation Model

The first of the two predictive frameworks developed in this study is a manually constructed linear equation model, grounded exclusively in the literature-derived factor weights established in Section 5.1. For each of the three outcome variables, a weighted linear combination of the normalised input factor scores is computed, producing a continuous composite risk score on a 0–1 scale. The general form of the linear prediction equation is as follows:

$$Risk\ Score = \sum (w_i \times x_i)$$

Where w_i represents the literature-derived weight of factor i for the outcome variable in question, and x_i represents the normalised survey response value for that factor. Respondents whose composite risk score exceeds a defined threshold — to be empirically determined during model calibration — are classified as high-risk, while those below the threshold are classified as low-risk.

This linear model serves a dual purpose within the study. As a standalone analytical tool, it operationalises the collective predictive understanding of the existing literature into a transparent, interpretable scoring framework. As a comparative baseline, it provides the reference point against which the data-driven weights produced by the machine learning model are evaluated, enabling a systematic, factor-level comparison between what the literature prescribes and what the data

supports.

5.3.2. Machine Learning Model

The second predictive framework employs two ensemble-based machine learning algorithms — XGBoost and Random Forest — trained on the SMOTE-augmented dataset. Both algorithms are selected for their demonstrated capacity to model complex, non-linear interactions among multiple co-varying input features, their robustness to class imbalance, and their established application in health-related predictive modelling contexts (Li et al., 2022).

The dataset will be partitioned using an 80/20 train-test split, wherein 80% of the augmented dataset is used for model training and hyperparameter optimisation, and the remaining 20% — supplemented by the 40 held-out real survey responses — is used for final model evaluation. To reduce the risk of overfitting and improve the reliability of performance estimates, k-fold cross-validation ($k = 5$) will be applied during the training phase.

Model performance will be evaluated using a combination of metrics selected to provide a comprehensive assessment of predictive accuracy across potentially imbalanced outcome classes. These include overall classification accuracy, the F1-score — which balances precision and recall and is particularly appropriate for imbalanced datasets — and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which measures the model's discriminative capacity across all classification thresholds. A minimum average accuracy threshold of 75% is established as the benchmark for acceptable model performance, consistent with benchmarks reported in comparable ML-based ocular health research (Li et al., 2022).

Upon training, the feature importance scores produced by both XGBoost and Random Forest — expressed as normalised weights on a 0–1 scale — are extracted and recorded for each input factor across each of the three outcome variables.

5.4. Comparative Analysis: Literature Weights vs. Machine Learning Weights

The final and most novel phase of the methodological pipeline involves a systematic, factor-level comparison between the literature-derived weights established in Section 5.1 and the data-driven feature importance scores produced by the machine learning models in Section 5.3.2. This comparison is conducted independently for each of the three outcome variables and for each individual input factor, producing a structured alignment matrix that documents the degree of convergence or divergence between the two weighting frameworks.

Three interpretive outcomes are possible for each factor. Where the literature weight and the ML-derived weight are in close alignment, this convergence is interpreted as cross-methodological validation — indicating that the existing empirical literature accurately captures the relative predictive importance of that factor within the context of this study's population and dataset. Where the ML-derived weight is substantially higher than the literature weight, this divergence suggests that the factor may be underrepresented or insufficiently examined in the existing literature, and warrants more targeted empirical investigation. Where the ML-derived weight is substantially lower than the literature weight, this may indicate either that the factor's effect is mediated or suppressed by other co-varying factors within the dataset, or that the synthetic dataset requires refinement through integration with a more comprehensive real-world data source.

The results of this comparison will be presented in a structured summary table alongside visualisations of weight distributions across both frameworks, providing a clear and accessible foundation for the interpretive conclusions of the study. Where systematic patterns of divergence are identified, specific recommendations for further research — whether directed at more granular empirical investigation of underweighted factors or at improvements to the data collection instrument — will be formulated and presented in the discussion section.

CONCLUSION

This research provides a critical exploration into the long-term ocular consequences of the global transition to a "digital-first" lifestyle. By moving beyond simple correlations and developing a predictive machine learning model, this study offers a method to quantify the hidden risks of screen-based consumption. The findings aim to demonstrate that myopia, astigmatism, and Computer Vision Syndrome are not merely inevitable byproducts of modern life, but are outcomes influenced by a measurable matrix of habits and environmental settings. While the reliance on a hybrid synthetic dataset is a limitation necessitated by current data privacy landscapes, the architectural success of the model provides a scalable blueprint for future healthcare applications. Ultimately, this thesis contributes to the field of health informatics by providing a tool that can alert users to high-threshold risks before transient symptoms evolve into permanent refractive errors, thereby advocating for more ergonomic and mindful digital engagement.

REFERENCES

- Ha, A., Lee, Y. J., Lee, M., Shim, S. R., & Kim, Y. K. (2025). Digital Screen Time and Myopia: A Systematic Review and Dose-Response Meta-Analysis. *JAMA network open*, 8(2), e2460026. <https://doi.org/10.1001/jamanetworkopen.2024.60026>
- Swetha, M., Sumitra, L. K., & Aniruddha, G. (2025). Effect of screen time on visual accommodation and eye fatigue. *International Journal of Academic Medicine and Pharmacy*, 7(5), 860–866. https://academicmed.org/Uploads/Volume7Issue5/164.%206129.%20JAMP_LK%20Sumitra_860-866.pdf
- Żuczek, O., Żuczek, A., Dyndał, K., Broda, M., Jędrzejewska-Rzezak, P., Urbańska, K., Szczap, I., Hawryluk, A., Marzec, K., & Mokrzycka, A. (2025). The impact of blue light from electronic devices on visual function, circadian rhythm, and wellbeing: A systematic review. *International Journal of Innovative Technologies in Social Science*, 3(47). [https://doi.org/10.31435/ijitss.3\(47\).2025.3904](https://doi.org/10.31435/ijitss.3(47).2025.3904)
- Zhang, N., Han, G., & Hao, R. (2026). The role of circadian rhythms in the pathogenesis of myopia. *Frontiers in Physiology*, 17, Article 1797489. <https://doi.org/10.3389/fphys.2026.1797489>
- Ece, B. Ş. D., & Altındaş, İ. (2025). Impact of screen use on tear breakup time: associations with ophthalmological factors and break frequency in office workers. *Arquivos brasileiros de oftalmologia*, 88(6), e20250083. <https://doi.org/10.5935/0004-2749.2025-0083>
- Carr, B. J., & Stell, W. K. (2020). The science behind myopia. Webvision. <https://www.webvision.pitt.edu/book/part-xvii-refractive-errors/the-science-behind-myopia-by-brittany-j-carr-and-william-k-stell/>
- Kaur, K., Gurnani, B., Nayak, S., Deori, N., Kaur, S., Jethani, J., Singh, D., Agarkar, S., Hussaindeen, J. R., Sukhija, J., & Mishra, D. (2022). Digital Eye Strain- A Comprehensive Review. *Ophthalmology and therapy*, 11(5), 1655–1680. <https://doi.org/10.1007/s40123-022-00540-9>
- Raziya, D., Sunitha, M., & Kumar, K. S. (2022). Association between screen time, blink rate, and symptoms of digital eye strain in adolescents: A cross-sectional observational study. *International Journal of Health Sciences and Research*, 15(1), 46–52.

<https://healthcare-bulletin.co.uk/article/association-between-screen-time-blink-rate-and-symptoms-of-digital-eye-strain-in-adolescents-a-cross-sectional-observational-study-4667/>

Shah, M., Natarajan, S. B., & Ahmad, N. (2025). Association of excessive screen time exposure with ocular changes leading to astigmatism in children. *PLoS ONE*, 20(4), Article e0317961. <https://doi.org/10.1371/journal.pone.0317961>

Harb, E. N., & Wildsoet, C. F. (2019). Origins of Refractive Errors: Environmental and Genetic Factors. *Annual review of vision science*, 5, 47–72. <https://doi.org/10.1146/annurev-vision-091718-015027>

Li, S. M., Ren, M. Y., Gan, J., Zhang, S. G., Kang, M. T., Li, H., Atchison, D. A., Rozema, J., Grzybowski, A., Wang, N., & Anyang Childhood Eye Study Group (2022). Machine Learning to Determine Risk Factors for Myopia Progression in Primary School Children: The Anyang Childhood Eye Study. *Ophthalmology and therapy*, 11(2), 573–585. <https://doi.org/10.1007/s40123-021-00450-2>

Goyal, M., & Mahmoud, Q. H. (2024). A Systematic Review of Synthetic Data Generation Techniques Using Generative AI. *Electronics*, 13(17), 3509. <https://doi.org/10.3390/electronics13173509>

Foreman, J., Salim, A. T., Praveen, A., Fong, D. W. K., Chia, S. S., Mushtaq, F. B., ... & Dirani, M. (2021). Association between digital smart device use and myopia: A systematic review and meta-analysis. *The Lancet Digital Health*, 3(12), e806–e818. [https://doi.org/10.1016/S2589-7500\(21\)00135-7](https://doi.org/10.1016/S2589-7500(21)00135-7)

Pezoulas, V., Zaridis, D., Mylona, E., Androutsos, C., Apostolidis, K., Tachos, N., & Fotiadis, D. (2024). Synthetic data generation methods in healthcare: A review on open-source tools and methods. *Computational and Structural Biotechnology Journal*, 23, 2892 - 2910. <https://doi.org/10.1016/j.csbj.2024.07.005>.

Chawla, N., Bowyer, K., Hall, L., & Kegelmeyer, W. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *ArXiv*, abs/1106.1813. <https://doi.org/10.1613/jair.953>.

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27. https://papers.nips.cc/paper_files/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html

Xu, L., Skoularidou, M., Cuesta-Infante, A., & Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. *Advances in Neural Information Processing Systems*, 32. <https://arxiv.org/abs/1907.00503>

Dillman, D. A., Smyth, J. D., & Christian, L. M. (2014). *Internet, phone, mail, and mixed-mode surveys: The tailored design method* (4th ed.). Wiley. <https://psycnet.apa.org/record/2014-34233-000>

Rosenfield, M. (2011). Computer vision syndrome: A review of ocular causes and

potential treatments. *Ophthalmic and Physiological Optics*, 31(5), 502–515. <https://doi.org/10.1111/j.1475-1313.2011.00834.x>

Hao, S., Han, W., Jiang, T., Li, Y., Wu, H., Zhong, C., Zhou, Z., & Tang, H. (2024). Synthetic Data in AI: Challenges, Applications, and Ethical Implications. *ArXiv, abs/2401.01629*. <https://doi.org/10.48550/arxiv.2401.01629>.

Huang, H. M., Chang, D. S. T., & Wu, P. C. (2015). The association between near work activities and myopia in children — a systematic review and meta-analysis. *PLOS ONE*, 10(10), e0140419. <https://doi.org/10.1371/journal.pone.0140419>

Sheppard, A. L., & Wolffsohn, J. S. (2018). Digital eye strain: Prevalence, measurement and amelioration. *BMJ Open Ophthalmology*, 3(1), e000146. <https://doi.org/10.1136/bmjophth-2018-000146>

Hiraoka, T., Miyata, K., & Oshika, T. (2018). Corneal irregular astigmatism and contrast sensitivity after refractive surgery. *Eye*, 32, 1839–1846. <https://doi.org/10.1038/s41433-018-0174-0>

Lanca, C., & Saw, S. M. (2020). The association between digital screen time and myopia: A systematic review. *Ophthalmic and Physiological Optics*, 40(2), 216–229. <https://doi.org/10.1111/opo.12657>

Talens-Estarellles, C., Cerviño, A., García-Lázaro, S., Fogelton, A., Sheppard, A., & Wolffsohn, J. S. (2023). The effects of breaks on digital eye strain, dry eye and binocular vision: Testing the 20-20-20 rule. *Contact Lens and Anterior Eye*, 46(2), 101744. <https://doi.org/10.1016/j.clae.2022.101744>

Wolffsohn, J. S., Calossi, A., Cho, P., Gifford, K., Jones, L., Jones, D., & Verkicharla, P. K. (2021). Global trends in myopia management attitudes and strategies in clinical practice. *Contact Lens and Anterior Eye*, 44(1), 101396. <https://doi.org/10.1016/j.clae.2020.10139>

Sheppard, A. L., & Wolffsohn, J. S. (2021). Digital eye strain: Advances in understanding and treatment. *Asia-Pacific Journal of Ophthalmology*, 10(3), 212–219. <https://doi.org/10.1097/APO.0000000000000385>

Ostrin, L. A., Sajjadi, A., & Benoit, J. S. (2021). Objectively measured light exposure during school and summer in children. *Optometry and Vision Science*, 98(10), 1126–1137. <https://doi.org/10.1097/OPX.0000000000001769>

Lawrenson, J. G., Shah, R., Downie, L. E., Graham-Cope, A., & Hull, C. C. (2021). Interventions for the reduction of myopia progression in children: An updated systematic review and meta-analysis. *Ophthalmic and Physiological Optics*, 43(1), 1–27. <https://doi.org/10.1111/opo.12920>

Ghorbani Mojarad, N., Williams, C., & Guggenheim, J. A. (2021). Near work, education, and myopia. *Taiwan Journal of Ophthalmology*, 11(3), 215–224. https://doi.org/10.4103/tjo.tjo_81_21

Tedja, M. S., Haarman, A. E. G., Meester-Smoor, M. A., Kaprio, J., Mackey, D. A., Guggenheim, J. A., & Klaver, C. C. W. (2021). IMI — Myopia genetics report. *Investigative Ophthalmology & Visual Science*, 60(3), M89–M105. <https://doi.org/10.1167/iovs.18-25965>

Shah, R. L., Guggenheim, J. A., & UK Biobank Eye and Vision Consortium. (2021). Genome-wide association studies for corneal and refractive astigmatism in UK Biobank demonstrate a shared role for myopia susceptibility loci. *Human Genetics*, 137(11), 881–896. <https://doi.org/10.1007/s00439-018-1942-8>

Xiong, S., Sankaridurg, P., Naduvilath, T., Zang, J., Zou, H., Zhu, J., & Xu, X. (2020). Time spent in outdoor activities in relation to myopia prevention and control: A meta-analysis and systematic review. *Acta Ophthalmologica*, 95(6), 551–566. <https://doi.org/10.1111/aos.13403>

APPENDICES

Appendix A - Initial Input and Target Factor Table

The following table represents the individual factors and targets with their labels and descriptions. As stated by the proposal, the individual factors are subject to change depending on further research understanding, experimenting and literature review. This table also acts as an indirect reference for the survey inputs.

Table 1.1. Model Input Factors

Feature	Description	Myopia (w)	CVS (w)	Astigmatism (w)
Daily Screen Time	Total daily screen exposure; directly drives near-work demand and eye strain	0.70	0.90	0.30
Continuous Usage Without Breaks	Uninterrupted screen use; prevents accommodative recovery	0.65	0.85	0.25
Intensity of Usage	Cognitive demand of tasks; high-intensity tasks increase accommodative load and reduce blink rate	0.60	0.80	0.20

Viewing Distance	Shorter distances increase vergence-accommodation demand and axial elongation risk	0.75	0.70	0.25
Ambient Lighting	Poor screen-to-environment contrast increases glare, strain and accommodative lag	0.55	0.75	0.20
Multiple Device Usage	Simultaneous multi-screen use extends near-work load and forces rapid refocusing	0.50	0.80	0.15
Blue Light Filters	Use of night mode / filters; reduces short-wavelength exposure and photophobic discomfort	0.35	0.50	0.10
Profession	Occupational near-work demand; higher screen-intensive roles correlate with greater strain and myopia prevalence	0.55	0.75	0.25

Outdoor Activity	Time in natural light; stimulates retinal dopamine release inhibiting axial elongation	0.80	0.45	0.20
Genetics	Parental prescription history; strongest hereditary predictor of both myopia and corneal irregularity	0.90	0.25	0.80
Age of First Prescription	Earlier onset correlates with greater progression risk; age also reduces accommodative flexibility	0.75	0.60	0.50
Gender	Females show slightly higher myopia prevalence and significantly higher CVS reporting rates	0.40	0.50	0.20

Appendix B - Survey Structure

Below is the initial survey structure used for gathering data. The overall structure of the survey stays largely the same over the course of the research process, with minimal changes made.

Q1. What is your age? (*Numeric input*)

Q2. What is your gender?

- Male
- Female

Q3. On a typical day, how many hours do you spend looking at digital screens in total? (phones, tablets, computers, TV)

- Less than 1 hour
- 1 to 2 hours
- 2 to 3 hours
- 3 to 4 hours
- 4 to 5 hours
- 5 to 6 hours
- 6 to 7 hours
- More than 7 hours

Q4. How long do you typically use a screen continuously before taking a break? (without looking away for at least 20 seconds)

- 10 to 20 minutes
- 20 to 30 minutes
- 30 to 40 minutes
- 40 to 60 minutes
- More than 60 minutes

Q5. How would you describe the typical intensity of your screen-based tasks?

- Low focus — occasional glances, brief checks
- Below average — standard scrolling and messaging
- Average — general content consumption with occasional reading
- Above average — frequent high-focus tasks, near-work required
- Very high intensity — demanding job with heavy digital content

Q6. How would you describe the lighting conditions in which you typically use your devices?

- Almost exclusively bright/high-brightness environments
- Mostly bright, occasional dim lighting
- Even mix of bright and dark environments
- Mostly dim, occasional bright environment
- Almost exclusively in dark/dim ambient lighting

Q7. How often do you use two or more screens simultaneously? (e.g. laptop + phone at the same time)

- Never (I focus on one screen only)
- Rarely (Only for quick checks and occasional information glances)
- Occasionally (Daily for short periods)
- Often (I use my phone and laptop often in conjunction)
- Almost always (my day-to-day tasks require me to use two or more devices simultaneously)

Q8. How far is your primary screen typically held from your eyes when in use?

- Very Close (Less than 20cm / 8 inches)
- Close (20cm to 35cm / Near work)
- Medium (35cm to 50cm / Arm's length)
- Standard (50cm to 70cm / Desktop monitor)
- Far (More than 70cm / Large TV or Projector)

Q9. How consistently do you use Blue Light Filters, "Night Shift," or "Night Mode" on your devices?

- Never
- Rarely (Only late at night)
- Occasionally
- Frequently (Most of the day)
- Always (Enabled on all devices at all times)

Q10. Which category best describes your daily professional or academic environment?

- No digital usage (Manual labor / Outdoor work)
- Low usage (Occasional check-ins / Retail)
- Moderate usage (Office work / Standard admin)
- High usage (Student / Extensive research)
- Extreme usage (Software Engineering / Data Analysis / Pro Gaming)

Q11. On average, how much time do you spend outdoors in natural light each day?

- Less than 30 minutes
- 30 minutes to 1 hour
- 1 to 2 hours
- 2 to 3 hours
- More than 3 hours

Q12. To your knowledge, do your biological parents have vision prescriptions / errors? (Nearsightedness / Astigmatism / Blurred Vision)

- No history / Neither parent
- One parent (Mild prescription)
- One parent (Strong prescription)
- Both parents (Mild to Moderate)
- Both parents (Strong prescriptions)

Q13. At what age do you remember first getting a prescription? Enter 0 if you have never had or currently do not have a prescription. (Numeric input)

Q14. What is the measure of your current nearsightedness prescription? If possible, an exact measure is preferred, or please select the measurement closest to the prescription you remember having. (Write the prescription in -. format, for example: -0.25)

- 0.0 to -0.5
- -0.5 to -1.5
- -1.5 to -3.0
- -3.0 to -4.5
- -4.5 and below
- Other

Q15. Over the last 2 years, how has your ability to see objects at a distance changed?

- Stable (No change in prescription or clarity)
- Worsened Slightly (Minor change / Slight blurriness for objects in the distance)
- Worsened Moderately (Noticeable blurriness / Need to focus hard to see distant objects)
- Worsened Significantly (Required much stronger glasses / Almost impossible to see far clearly)

Q16. How often do you experience physical symptoms such as headaches, eye-strain and similar?

- Never
- Rarely (Once or twice a month)
- Occasionally (Once or twice a week)
- Frequently (3 to 5 times a week)
- Almost Daily

Q17. When looking at text, bright lights or specifically lights at night, how often do you see "ghosting," shadows, or double-edges?

- Never (Clear edges)
- Rarely (Only when eyes are very tired)
- Occasionally (In low-light conditions)
- Frequently (Noticeable on most text)
- Constantly (Persistent distortion regardless of light)